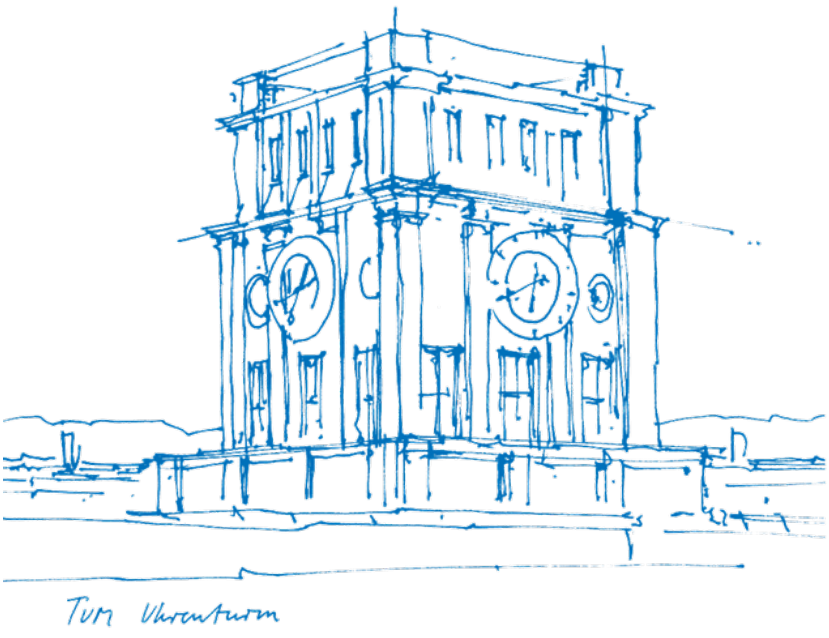
Chair of Information Systems and Business Process Management (i17)
Department of Computer Science
TUM School of Computation, Information and Technology
Technical University of Munich

Master's Thesis in Information Systems

Xin Wen

Document-Based Process Modeler



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Dokumentatenbasierte Bearbeitung und Änderung von Prozessmodellen

Thesis for the Attainment of the Degree
Master of Science

at the TUM School of Computation, Information and Technology,
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Chair of Information Systems and Business Process Management (i17)

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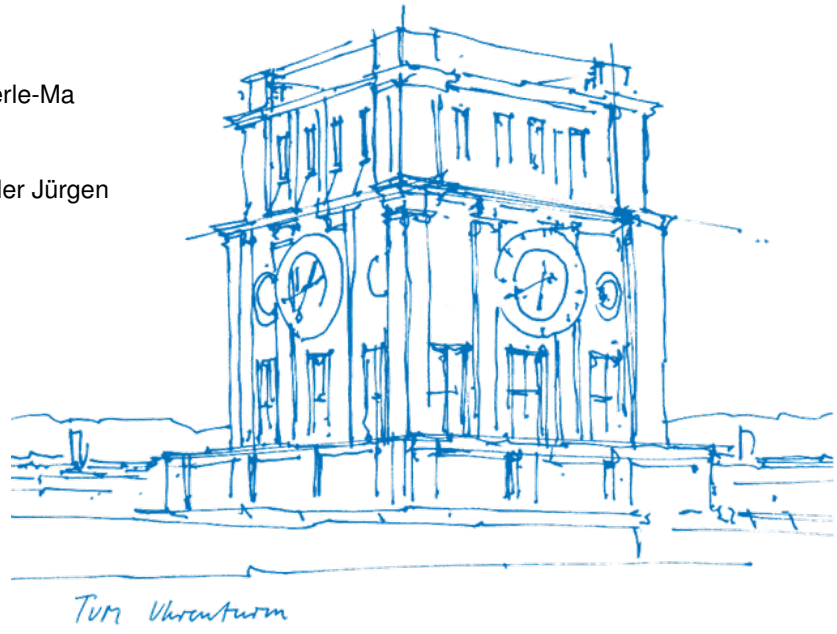

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Abstract

Business processes are valuable assets of organizations, yet most process knowledge is stored in unstructured formats such as Word documents and PDF files. Manual transformation from unstructured documents into structured process models is time-consuming and error-prone. Large language models (LLMs) have proven effective in supporting efficient process modeling, but they cannot fully replace human experts, especially in domain-specific scenarios. Hybrid human-LLM process modeling is therefore a promising approach to leverage the strengths of both sides and compensate for their respective weaknesses. However, existing tools and approaches still focus on model generation capabilities and quality, making it still inconvenient for the use scenario of creating process models from documents. This work addresses these limitations by proposing and developing an integrated web-based system that supports document-driven process modeling with LLMs through a more seamless, efficient, and user-friendly workflow. The system unifies document management, direct text selection, automated model generation, visual traceability between text and models, subprocess relationship establishment, and model version control within a single workspace. A user evaluation with four process modeling experts shows that the system is intuitive, usable, and valuable for multi-model and document-intensive scenarios.

Keywords: *Business Process Management, Process Modeling*

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Introduction

Motivation

Business processes represent core operational assets of modern organizations, defining standardized workflows, responsibilities, and execution logic for daily operations. However, a large proportion of enterprise process knowledge remains trapped in unstructured document formats, such as Word, PDF, plain text, and internal manuals. Manually converting such unstructured textual content into formal, executable process models is labor-intensive, time-consuming, and prone to human error, reducing the efficiency of process design and digitization[1].

In the era of generative artificial intelligence (AI), large language models (LLMs) have emerged as powerful enablers for automated process modeling, demonstrating strong capabilities in understanding natural language, extracting logical structures, and generating standardized process notations. Nevertheless, LLMs cannot fully replace human experts, especially in domain-specific scenarios requiring professional knowledge, contextual judgment, and structural decision-making[2]. Therefore, hybrid human-LLM process modeling has become a promising paradigm that combines human domain expertise with AI efficiency.

Despite rapid advancements, existing LLM-assisted process modeling solutions suffer from critical limitations in document-driven scenarios. Most tools adopt a "single-view, single-model" paradigm, which lacks scalability for complex systems involving multiple processes and subprocesses. To overcome these gaps, this thesis proposes an integrated tool, Document-Based Process Modeler. The tool enables a seamless, efficient, and user-centric modeling workflow with three core innovations: unified workspace management for correlated documents and models; direct text selection from uploaded documents as LLM input; and side-by-side visualization to establish explicit visual link between source text and generated process model to make a more intuitive and verifiable modeling experience.

Research Questions

This thesis addresses the following research questions:

RQ1 How do existing tools support AI-assisted process modeling multiple documents as input?

RQ2: What functionalities should a system provide to support AI-assisted process modeling from multiple documents, and how should these functionalities be designed and integrated in a seamless and user-friendly manner?

RQ3 How do user perceive the benefits and limitation of the system?

Contribution

The contribution of this thesis consists of the following aspects:

Firstly, a systematic review of the existing tools and approaches for AI-assisted process modeling is conducted, which provides insights into the current state of the art and guides the design of the proposed system.

Secondly, an integrated system is designed and developed to support process modeling from documents using LLMs. The system offers a more seamless, efficient, and user-friendly workflow by integrating multiple functionalities, such as document management, model generation, and traceability between text and model.

Thirdly, the proposed system is evaluated through a user study with process modeling experts. The evaluation results provide insights into the system's benefits and limitations from the users' perspective, which can inform future improvements and further research in this area.

Methodology

The research process of this thesis follows the three-cycle view of Design Science Research (DSR) proposed by Hevner [3]. This framework consists of three interconnected cycles: the relevance cycle, the design cycle, and the rigor cycle, which link the problem environment, artifact development, and the knowledge base. In this work, these cycles are applied as follows.

The relevance cycle is initiated by problems in current process modeling tools, especially the lack of support for document-based modeling. These problems are continuously discussed with modeling experts throughout the development process to identify practical needs and define system requirements.

The design cycle focuses on designing, implementing, and improving the artifact, the Document-Based Process Modeler. The system is developed and evaluated iteratively. Feedback from the evaluation (RQ3) is used to refine and improve the design.

The rigor cycle is supported by continuously reviewing state-of-the-art AI-assisted process modeling tools in the literature review (RQ1), ensuring that the design is grounded in prior work and contributes to the research field.

Evaluation

The evaluation of the proposed system is done with 4 experts in qualitative way. It consisted of task-based activities and a follow-up questionnaire to collect user feedback on the tool's usability and perceived usefulness.

Structure

The remainder of this thesis is structured as follows:

Chapter 2 reviews the related work of AI-assisted process modeling tools. Chapter 3 presents the design solution of the tool, including its requirements, architecture, and functionalities. Chapter 4 describes its technical implementation. Chapter 5 describes the evaluation of the proposed system, including the methodology, results. Chapter 6 discusses the implications of the findings and potential future work. Finally, Chapter 7 summarizes the current work and outlines future work.

Related Work

Background

Business Process Management (BPM) refers to the systematic design, analysis, and optimization of organizational processes to improve efficiency and effectiveness, while Business Process Modeling (BPM) focuses on representing these processes in a structured and understandable form. BPM languages can be broadly categorized into conceptual, formal, and executable types. Conceptual languages, such as Business Process Model and Notation (BPMN), provide intuitive graphical notations that facilitate communication between business and technical stakeholders, although they

lack fully formal semantics [4]. In contrast, formal languages like Petri Net offer mathematically precise representations, enabling rigorous analysis of properties such as correctness and deadlock-freedom [5]. To operationalize process models, executable structures such as Refined Process Structure Tree (RPST) are used to transform models into well-structured, executable components [6].

The generation of process models from textual descriptions, known as text-to-model, has evolved significantly over time. Early approaches relied on rule-based methods within Natural Language Processing, mapping linguistic elements such as verbs and nouns to process activities and actors [7]. Subsequent statistical and machine learning approaches improved robustness but required annotated data and extensive feature engineering. More recently, advances in Large Language Models have enabled new approaches to process model generation from natural language, with studies indicating their potential to streamline modeling tasks and reduce manual effort, while still requiring evaluation of the generated models [1], [8].

AI-Assisted Process Modeling Tools

While many tools exist that deal with analysis of large document sets, to the best of our knowledge no integrated tool with the dedicated purpose of creating Process Models from documents set exists. Therefore here we will explore some related tools.

In total, eleven AI-assist modeling tools are explored during the literature review (See table 2). Here, we mainly focus on the key approaches and the distinguishing features they offer.

In the 12 hits of the first search query("allintitle: process modeling generative AI"), the tool ProMoAI [9] is identified. It is the first tool shown in the academic literature to leverage LLMs for automated process model generation, introduced in 2024, and is frequently referenced in subsequent work. ProMoAI enables the transformation of natural language into process models and supports iterative refinement through feedback loops, demonstrating the feasibility of LLM-based process modeling.

In the second query ("allintitle: process modeling LLM"), among 16 hits, 5 tools are identified. The first two noticed is BPMN-Chatbot [10] and BPMN Assist [11]. BPMN-Chatbot is an early one in 2024 with key strength in toke efficiency[12], compared with PromoAI's approach. Its method was introduced in. BPMN Assist [11] is a very latest one. BPMN Assistant [11] represents a more

Table 1
Literature review: horizontal research

Keywords for search	*hits*	*selected*	*Selection criteria*
allintitle: process modeling generative AI	12	1	Process modeling tool
allintitle: process modeling LLM[s]	16+7	5+0	Process modeling tool
allintitle: process modeling large language	49	1	Process modeling tool
allintitle: conversational process modeling	8	1	Process modeling tool
allintitle: BPMN generative AI	2	0	Process modeling tool
allintitle: BPMN LLM[s]	12+8	2+1	Process modeling tool
allintitle: BPMN large language	7	1	Process modeling tool
Results horizontal search	121	11	
Results vertical/backward search			
Added papers		1	
Overall:		12	

Source: https://scholar.google.com/

recent development, emphasizing structured intermediate representations (e.g., JSON) to improve reliability, editing capabilities, and performance in interactive modeling scenarios.

The tool KICoPro is introduced in [13], focusing on human-in-the-loop process modeling, where users collaborate with AI systems during model creation.

The tool HyperMode is introduced in [14], which combines LLM-based generation with a direct manipulation interface, allowing users to refine process models interactively by selecting and modifying elements within the diagram.

Additionally, the tool BPMNGen is firstly introduced in [15] then further developed and evaluated in [16]. It is a different one using NLP, with empirical evaluations showing strong semantic quality for simple to moderately complex processes.

In the last search attempt for term "process modeling" by combining it with "large language", one more tool PRODIGY is explored mentioned within the paper [17]

In the search query ("allintitle: conversational process modeling"), the only tool AutoBPMN.AI [18] is found.

Finally, combining “BPMN” with "generative AI", "LLM[s]", or "large language" led to the identification of 3 additional tools, including BPMN-Chatbot++ [19], LLM4BPMNGen [20], and Nala2BPMN [21].

BPMN-Chatbot++[19] is found in Tool no name [22]

LLM4BPMNGen[20]

Nala2BPMN [21]

Table 2
Related work on AI-assisted process modeling tools (simplified)

Tool	Year	Input	Output	Key Contribution
ProMoAI	2024	text	BPMN/PNML	Code-based generation with correctness focus
BPMN-Chatbot	2024	Text/voice	BPMN	Conversational modeling with high token efficiency
Nala2BPMN	2024	text	BPMN	Hybrid LLM + algorithm pipeline
PRODIGY	2024	text	BPMN (text)	RAG-based enterprise modeling
LLM4BPMNGen	2025	Audio/text/voice	BPMN	Audio upload + structured prompting
BPMN-Chatbot++	2025	Text/voice	BPMN	Improved prompting and interaction
AutoBPMN.AI	2025	text	CPEE Tree	Executable model generation
HyperMode	2025	text	BPMN	Direct manipulation + LLM interaction
BPMNGen	2026	text	BPMN	Conversational generation + human-centered evaluation
BPMN Assist	2026	text	BPMN	JSON-based intermediate for efficient editing
KICoPro	2026	text	BPMN	Able to answer questions and explain descisions

AutoBPMN.AI

Here, we provide a more detailed in depth review of the AutoBPMN.AI tool, as we will leverage its LLM service for the automated process model generation in our proposed system.

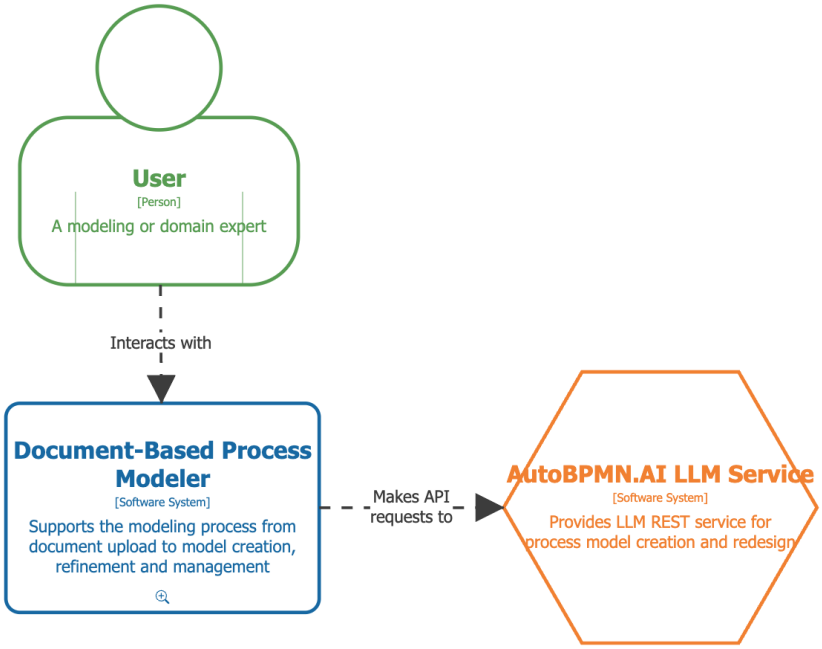
Solution Design

Requirements

Context and Goal

First, the context and objectives of the solution design are defined. The system is provided as a web-based application, as illustrated in Figure 1. The target users include both process modeling experts and domain experts who need to create process models from textual documents. To improve modeling efficiency and support non-expert users, the system leverages the LLM service of AutoBPMN.AI for automated model generation. The overall objective is to provide an integrated platform in which users can complete the entire modeling workflow—from document upload to model creation, modification, and subsequent management—in a seamless and user-friendly manner, thereby addressing RQ2.

Figure 1
System context



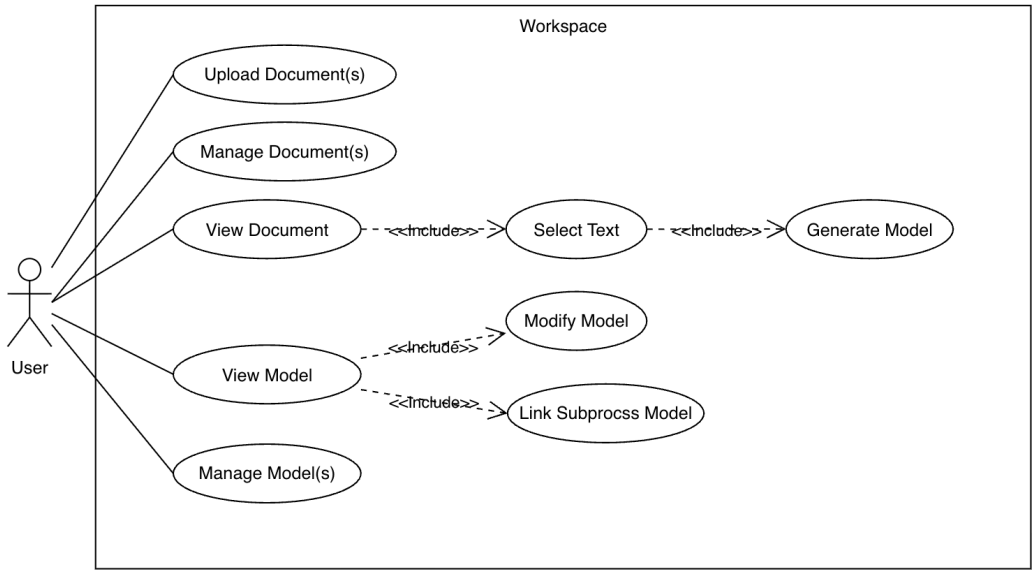
Requirements Elicitation

Functional requirements define the system's expected behavior by describing the functions, inputs, outputs, and interactions that enable the system to fulfill its intended objectives [23]. In this work, we

adopt the classification proposed by [24], further diving the functional requirements into minimal requirements, which represent the indispensable core functionalities required for the system to operate, and enhancing requirements, which include additional features that improve usability and overall performance but are not strictly necessary for basic operation.

When starting the work, the following minimal requirements for the system have been elicitated by my supervisor. Firstly, the system should allow users to upload multiple documents. The basic process modelling is defined as follows: first, the user should be able to select multiple passages from a single document; then, by clicking a Generate button, the system will send the selected text to the LLM service. The system should then display the generated model side-by-side with the document to facilitate verification. Besides, the system should always provide a clear visual link between the text and the generated model. Then a user should be able to select other passages to create more models. Another important requirement is that the system should allow users to establish process-subprocess relationships between models. This flow should be designed on a single web page to avoid users having to switch between tools. The activities described are depicted in the Figure 2. Here, we use the concept of workspace for this dedicated interface, a term commonly used across many software tools, where the modeling flow is performed.

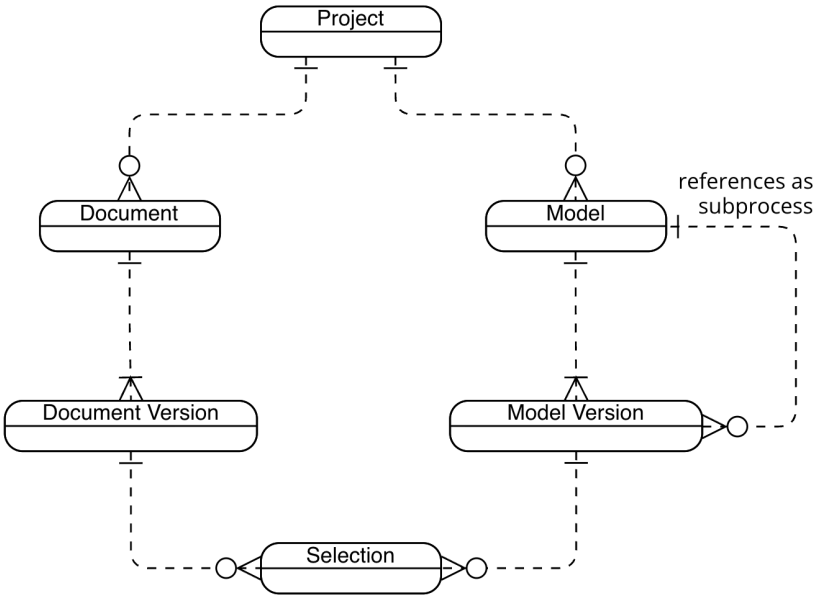
Figure 2
Usage model



Regarding non-functional requirements, only usability, as a very general goal, is defined at this stage. Besides this, a series of technical design constraints are specified. In addition to the two constraints already introduced in the system context, i.e. the system should be implemented as a web-based application and should leverage AutoBPMN.AI's LLM service, two more constraints are explicitly defined. First, the frontend should be developed using native web technologies (JavaScript, HTML, and CSS). Second, the visualization of the process model should be implemented using the CPEE Layout[1] library, following the same layout as used in AutoBPMN.AI, which makes the models easy to export and import, and also provides the potential for them to be uploaded to the CPEE environment without additional manual effort, where they can be directly further manipulated and executed.

With the development, more enhancing requirments are elicitated. Firstly, we use the term "Project" for

Figure 3
Conceptual data model



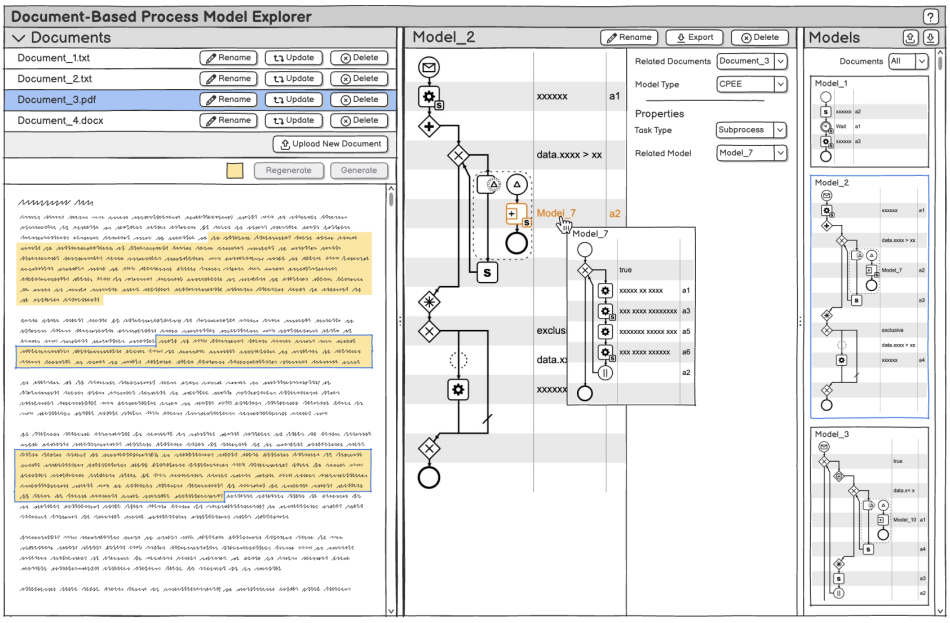
User Interface and Interaction Design

The layout design of the main workspace is shown in the figure 4 in a 3-pane layout. The left pane is for document uploading and viewing, the middle pane is for process modeling and the right pane is

[1] `https://github.com/etm/cpee-layout`, last access 2026-04-17

for process models viewing and project graph visualization. The details are divided as the following: a) Document List b) Document Viewer c) Model Editor d) Model List e) Project Graph.

Figure 4
Workspace UI Layout

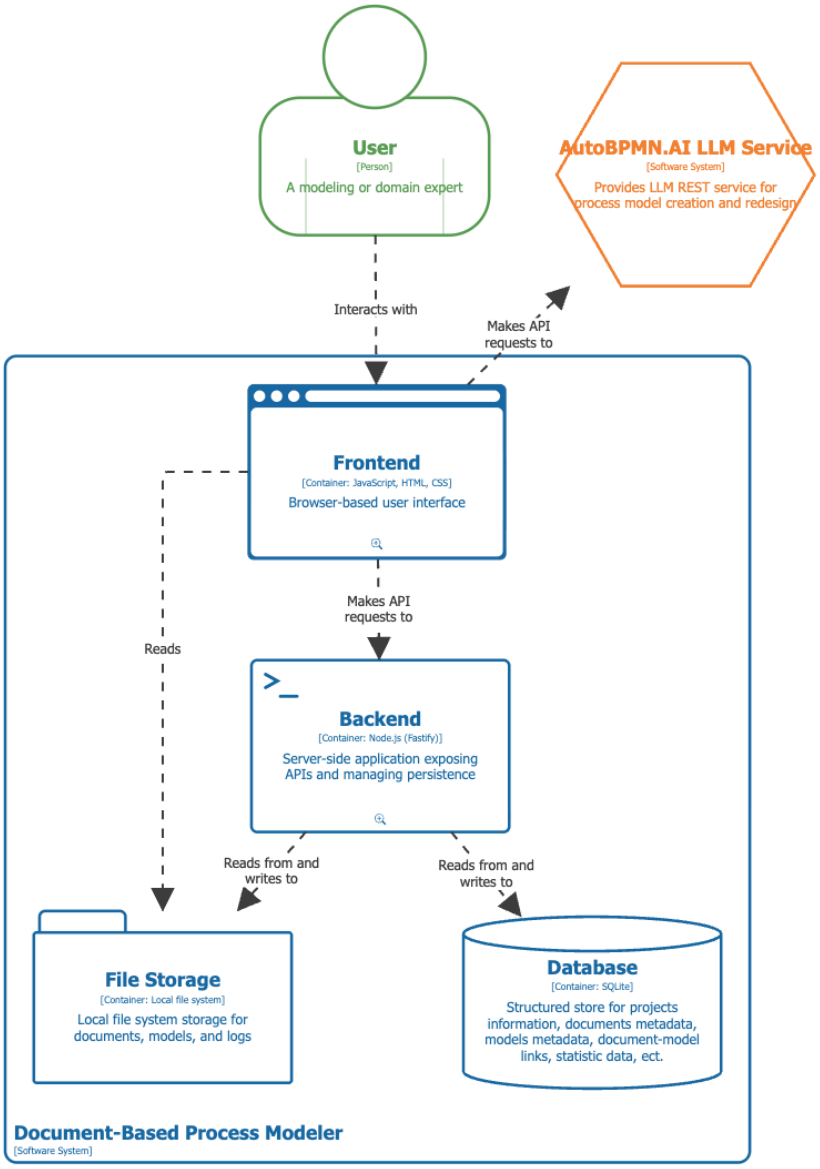


The user interface layout of the workspace page.

System Architecture

The system architecture is illustrated in Figure 5. It follows a client–server architecture consisting of a frontend, a backend, and a persistence layer, with integration of an external LLM service. The frontend is implemented as a multi-page application (MPA), consisting of a home page, a workspace page, and a statistics page, where the workspace page represents the core of the system, supporting document handling and process model generation. For LLM-based model generation, the frontend sends requests to the external LLM service directly, and the generated results are returned to the frontend for user review and interaction. After user confirmation, the data is transmitted to the backend for persistence. The frontend communicates with the backend via RESTful APIs, through which the backend manages projects, documents, and models and handles data storage. The persistence layer follows a hybrid approach: the actual content of documents and models is stored in the local file system, while their metadata (e.g., version numbers and names) and application data, such as project information and relationships between process models, are maintained in a database.

Figure 5
System architecture



The system architecture of the document-based process modeling tool.

Implementation

Frontend

Architecture

The frontend was implemented as a native web application using JavaScript, HTML, and CSS. Since the main modelling workflow is performed in the workspace page, only the workspace architecture is described here. The workspace follows a layered structure: UI modules handle rendering and user interaction, service modules coordinate the workflow logic, and store modules keep the client-side state synchronized across the document viewer, model editor, and project graph.

The workspace initialization logic validates the current project, loads the required UI modules, and triggers the initial data loading. At runtime, the workspace service loads document, model, and subprocess metadata, initializes the stores, and selects an initial document if available. Later actions, such as opening a version, generating a model, or navigating to a subprocess, are handled through the same orchestration flow.

State Management

The frontend uses a lightweight publish-subscribe mechanism to coordinate the document list, document viewer, model editor, and project graph. The base implementation, shown in Figure 6, stores local state and notifies subscribers through patch objects.

This mechanism is extended through specialized stores for entity collections, versioned entities, workspace state, document-viewer state, model-editor state, and the project graph. The approach was chosen because it supports synchronized updates across several panes without introducing a larger frontend framework.

Figure 6
store implementation

```
export class Store {
  constructor(initialState = {}) {
    this.state = { ...initialState };
    this.subscribers = new Set();
  }

  subscribe(fn) {
    this.subscribers.add(fn);
    return () => this.subscribers.delete(fn);
  }

  notify(patch) {
    this.subscribers.forEach((fn) => fn(patch));
  }
}
```

Document Processing

Document upload is partly handled on the frontend so that heterogeneous source formats can be normalized before entering the main interaction flow. The implementation accepts PDF, DOC, DOCX, and plain-text input. Word documents are converted directly to HTML, while PDF files are processed into a structured HTML representation.

For PDFs, the frontend extracts text items, groups them into lines, classifies them into headings, lists, and paragraphs, and renders the result as semantic HTML. This normalization is necessary because later operations, especially selection serialization, overlay rendering, and re-anchoring, depend on stable DOM text ranges rather than on raw file content.

Text Selection Rendering

One of the central frontend mechanisms is the transformation of browser text selections into persistent, version-aware linkable objects. A user selection is serialized into a text-position selector together with a text-quote selector, following the anchoring idea of the W3C Web Annotation Model

[25]. The text-position selector identifies the range in the current text stream, while the text quote preserves the selected wording for later validation or re-anchoring.

Figure 7 shows the core implementation used to convert a browser range into a persistent selector object.

Figure 7
Core implementation of selection serialization

```
export function serializeRange(range) {
  const root = getRoot();
  const textPosition = textPos.fromRange(root, range);
  const textQuoteSelector = textQuote.fromTextPosition(root, textPosition
      );

  return {
    textPosition,
    textQuote: textQuoteSelector,
  };
}
```

When a stored selection is displayed again, the frontend resolves the serialized selector back into a DOM range. This allows existing links and older model versions to be visualized in the document viewer.

Virtual Overlay Strategy. Selection rendering is implemented as a virtual overlay rather than by wrapping the original text nodes. The selection module collects client rectangles, filters and de-duplicates them, removes oversized container rectangles, and reconstructs more accurate line-level rectangles when necessary. The resulting geometry is then rendered as overlay elements above the document content. This avoids mutating the source DOM and keeps selection rendering stable across re-renders, updates, and version switches.

Link Update Detection. The frontend also distinguishes between semantic changes in a linked selection and purely visual updates. Link drafts are classified into three states: no change, color-only change, and text-changed. This distinction is important during document-version updates,

where a visual modification should not invalidate the semantic relation between a passage and a model.

Modeling

The modeling flow coordinates the document viewer, the model service, and the model editor state. Selected text is aggregated, optionally combined with an additional prompt, and sent to the LLM-backed generation endpoint. The returned XML model data is then loaded into the model editor store and rendered in the editing pane.

The frontend also keeps the relation to the originating document version so that model editing remains synchronized with the document context. This logic supports new models, regenerations, refinements, and historical versions.

Rendering Customization

Instead of implementing a process-model editor from scratch, the frontend reuses the CPEE layout and rendering infrastructure and extends it where necessary. The main customization is the workflow adaptor layer, which dispatches additional events for call and subprocess nodes.

These events connect the generic renderer to workspace-specific behavior such as subprocess navigation and contextual interaction. Together, the layered frontend organization, pub-sub stores, document normalization, overlay rendering, and renderer customization support the end-to-end modelling workflow.

Backend

Architecture

The backend is implemented using a layered architecture. In this system, the layered structure is realized through route, service, and repository layers, each with a clearly separated responsibility. Route modules expose REST endpoints, service modules implement the application logic, and repository modules encapsulate data-access operations. Fastify.js was chosen as the backend framework because it is lightweight and provides native support for schema-based validation of requests and responses.

The schema layer is important because the system exchanges structured selection data between frontend and backend. As shown in Figure 8, selectors are validated before entering the service logic. The same approach is used across document, model, and link endpoints.

Figure 8
Example JSON schema used in the backend for text quotes

```
const textQuoteSchema = {
  type: "object",
  required: ["exact"],
  additionalProperties: false,
  properties: {
    exact: { type: "string" },
    prefix: { type: "string" },
    suffix: { type: "string" },
  },
};
```

The backend combines relational persistence with file-based storage. Metadata such as projects, documents, models, versions, and links are stored in SQLite, while large document contents and model XML payloads are stored as files in the persistence directory. This hybrid design keeps metadata queryable without placing large HTML and XML payloads directly in the database.

The service layer implements the application-specific workflows used by the frontend. For example, when a new document version is created, the backend stores the content, updates the latest-version pointer, and copies the latest link information. When a new model or model version is created, the backend persists the XML payload, updates version references, stores the related link, and records monitoring information. The backend also handles cross-cutting concerns such as soft deletion, logging, and aggregated workspace endpoints.

Database Schema

Firstly, Project, Document, Model and Document Model Link are the minimal tables needed to support the proposed system. It is also how the system was implemented when it first adopted the database. The Document and Model tables were used to store metadata like name, words count of

a document. Theoretically, the Document Model Link table can be stored as an attribute within the model version, since the system is currently designed to allow one model to be derived from only one document. But it is a decision made very early to store it in a separate table to easily support a potentially many-to-many (M2M) relationship in the future.

The Model Version table is added when the model versioning feature is about to be implemented. The Document Version table is added at the same time to prepare for supporting the Document Update feature, as both tables follow a similar logic at the DB operation level.

The Subprocess Link table is an additional table used to support the Project Graph. As this relationship information has been embedded in Model Data, the model visualization itself does not need this data. However, for the project graph, the relationship is hard to be parsed from the model data; therefore, such a table is designed to easily retrieve the process-subprocess relationship.

The Selection and SelectionHistory tables are then added when designing the Document Update feature. Before the selection data (e.g., color, position) for a Document Model Link is stored as an attribute in the format of a stringified JSON object. To better support auto-reanchoring for document upload, Selection and Selection History are designed as separate tables. The separate Selection table can make selection data easier to retrieve for reanchor operations. Besides, a Selection History is designed to make it easier to track the quality of reanchor.

Finally, the GenerationAttempt and Version Event tables are the two tables designed to monitor the system behavior. The Generation Attempt table is used to monitor the accept/decline decision for the LLM-generated output. The Version Event is mainly to store the manual edit behavior or a system auto-update behavior, e.g., selection reanchor for document update.

Evaluation

Evaluation Procedure

The developed system was evaluated with four modeling experts to assess its usability and perceived usefulness, as well as to gather feedback on potential improvements and additional features. The

evaluation process included a short demonstration, task-based activities, and a questionnaire. Each evaluation session lasted approximately 40 minutes.

The session was structured as follows: It began with a demonstration to introduce the tool's end-to-end workflow and key features. After that, the participants were asked to complete two tasks. The first was an exploratory task based on a fictional B2B order-processing scenario, consisting of two wiki-style documents[2] in DOCX format. It was designed to help participants familiarize themselves with the system. The second was a more realistic process modeling task based on an open-source SOP document[3] in PDF format. For both tasks, participants were not given strict instructions on how to proceed, allowing them to explore the system freely. Due to the known issue of the PDF file rendering, the link to the online version was provided. If time permitted, participants could additionally use the system with their own documents.

Following task completion, participants completed a questionnaire including the following open-ended questions:
(1) What aspects of the tool did you like while using it?
(2) What aspects of the tool did you not like or find problematic?
(3) What improvements or additional features would you most like to see in the tool?
(4) If you needed to model processes from documents in the future, would you consider using this tool? Why or why not?

Evaluation Results

Overall Perception and Task Completion

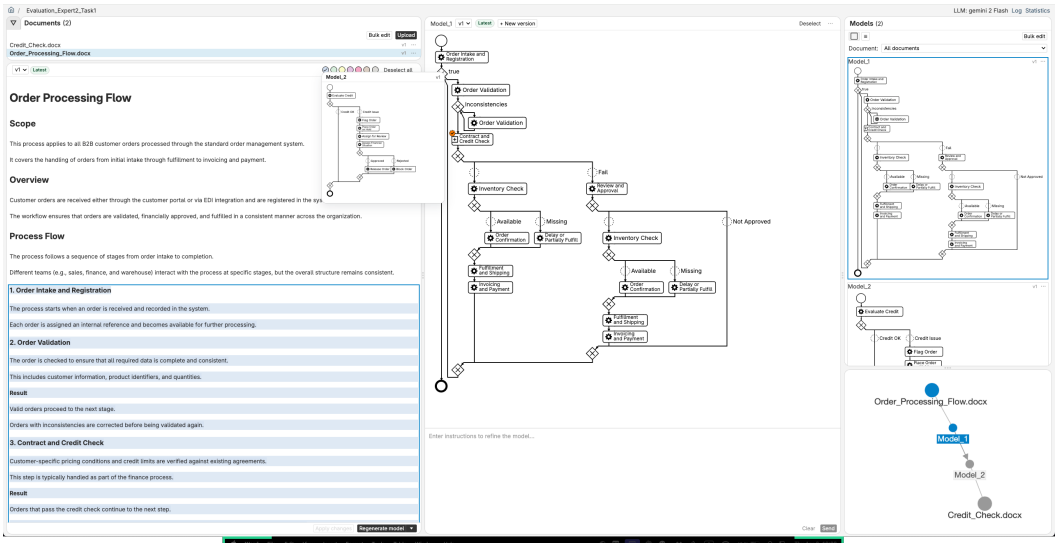
Overall, participants perceived the tool as usable and intuitive. Figure 9 and Figure 10 show the modeling results from one expert for the first and second tasks, respectively.

One interesting observation is that while most participants tended to start with, or at least try once, selecting larger portions or all of the text when reading a new document, one participant began by creating models from smaller segments. These different strategies are illustrated in Figure 11.

[2] https://github.com/ga94hor/doc-based-process-modeler/tree/evaluation_260324/evaluation_data/Task1_Fictional_B2BOrderProcessing

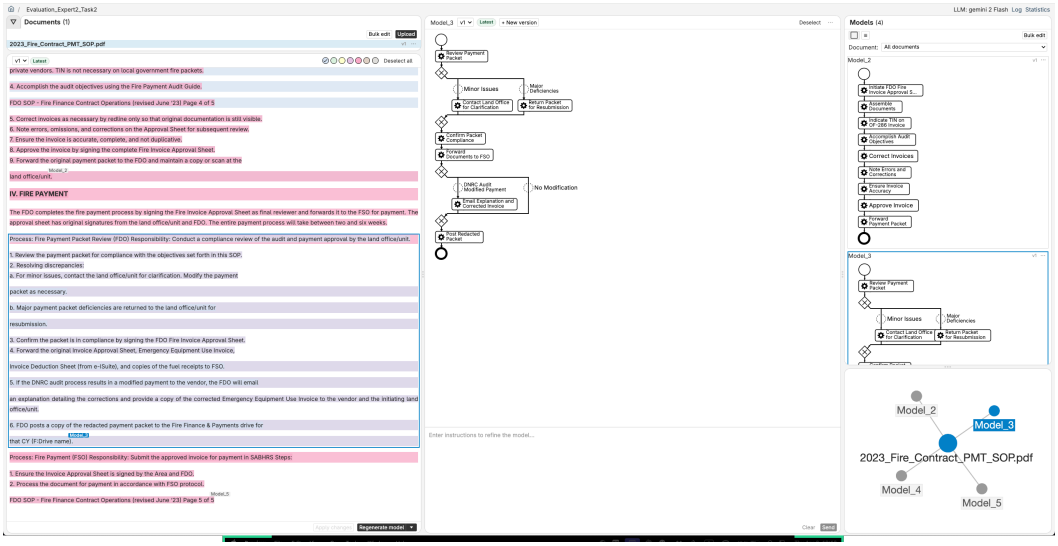
[3] https://dnrc.mt.gov/_docs/forestry/Wildfire/agreements-plans-guides/DNRC-Manuals/300-Fire-Business-Manual/Appendix-300-Manual/2023_Fire_Contract_PMT_SOP.pdf

Figure 9
Modeling result



Modeling result from an expert for the first task.

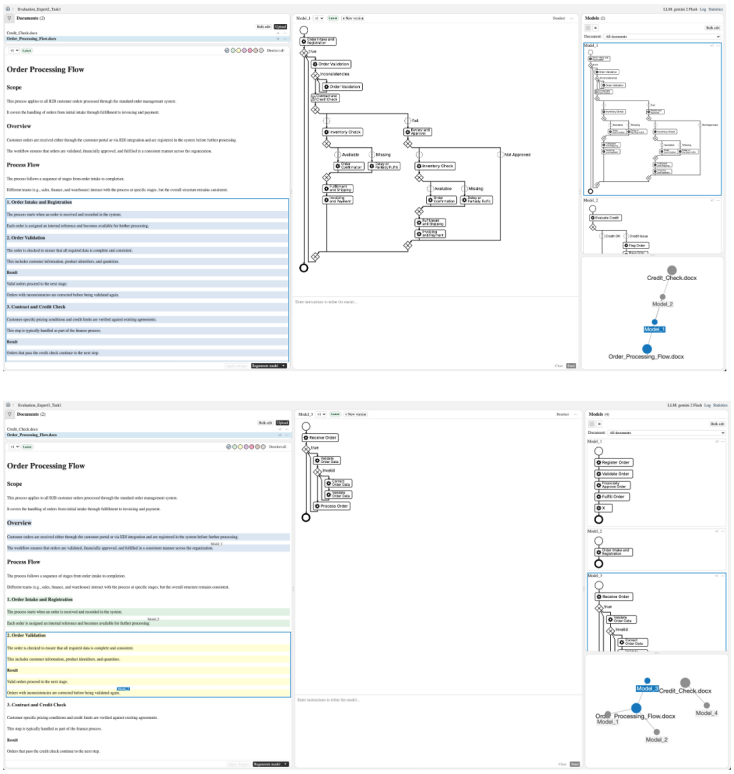
Figure 10
Modeling result



Modeling result from an expert for the second task.

For detailed task completion and feedback, the participants showed varied interests and offered differing perspectives on the system. The result on each feature is summarized in Table 3, and will be described in detail in the following sections.

Figure 11
Different modeling strategies



Different modeling strategies observed during task execution (above: larger part selection; below: step-by-step selection)

Positive Aspects

Firstly, the following aspects are generally perceived positively by the participants: the performance of the integrated LLM service, the easy-to-establish process-subprocess relationship, and model version control.

Furthermore, several additional positive aspects were mentioned. First, the embedding of relevant information, such as selected text and prompt history, within the model's XML data is considered a well-integrated design feature. Secondly, logging and statistical information elicited interest among some participants due to its potential for further analysis. Thirdly, the project graph visualization was mentioned as helpful, especially for larger systems.

Problems in Interaction and Use

The following outlines the main problems and challenges reported by participants, as well as notices observed during the evaluation.

Table 3
Summary of Evaluation Results by Functionality

Functionality	Positive Aspects		Problems and Limitations	
	Item	**n**	**Item**	**n**
Document Rendering and Interaction			Readability issue	1
			Expected AI-assisted search capability	1
Modeling Process			Unintuitive to start a new modeling by deselecting the current model	2
			Unnecessary review step (of the initial generated model)	2
Integrated LLM Service	Fast	2	Lack of transparency in the conversational modeling logic	2
	Good quality	2		
Model Versioning	Helpful	3	Expected version comparison	1
Augmented Model Data	Well-considered	2		
Text-Model Link	Clear	1	Expected fine-grained traceability	2
Process-Subprocess Relationship	Convenient	3		
Project Graph	Helpful for larger systems	1	Unclear node types and version representations	2
			Expected more expressive visualization	1
Log and Statistics	Potentially useful for analysis	1		

n = number of participants who mentioned each item

First, for the document's rendering and interaction, only one feedback regarding the readability issue of PDF documents was mentioned in feedback. But it is observed that one participant started with reading the original PDF online firstly and one participant tried to use browser-extension to search for keywords.

Regarding the modeling process design, two problems were mentioned. First, the step to start a new modeling by a explicit deselection step of the current model was mentioned. Two reply received in the feedback, but it is a feature generally perceived as not intuitive immediately and needs additional clarification. Second, due to the sufficient good quality of the initial model, the review step was

mentioned by one participant as totally unnecessary as there is version control and by another one as the initial generated model was mentioned as an unnecessary step.

Another main problem mentioned and observed is that the system's conventional modeling logic is not transparent to users and needs additional explanation. While the initial model generation from text was relatively clear, the model modification process was less transparent. They did not understand how their inputs were processed or how the system used the selected text and existing model.

Potential Improvements and Additional Features

Besides the problems found during system use, many valuable insights were collected.

For the existing features, improvements to the project graph visualization were mentioned most often. This includes clearer differentiation and identification of elements (i.e., what type of element a node is and which version it represents), as well as using more expressive visual designs, such as size, color saturation, and distance, to show additional information beyond type.

For document-related features, an AI-assisted document search and interaction function was suggested. This would allow users to query the document by providing an instruction, after which the retrieved parts can be automatically selected and used directly or as a basis for further adjustment for model generation.

For the process modeling features, one notable desired improvement emphasized by the participants is the need for finer-grained traceability between source text and model. When selecting larger text passages, only a subpart of the text is actually used and reflected in the generated model, and it is unclear which parts are actually used. Therefore, a clearer indication of the actual text used for model generation is required, ideally at the level of individual tasks. Secondly, model comparison was suggested as a valuable feature to help users understand and verify changes between different model versions, making it easier to identify modifications and track model iterations over time. Beyond this, participants also expected more general advanced AI modeling capabilities, including explainability of the generated model through conversational interaction with chatbots, as well as support for model quality evaluation and automatic improvement.

Future Use Intention

Three participants indicated an intention to use the system in scenarios involving the creation of multiple models. In such cases, the system's features, such as text-model link, process-subprocess relationship, and model management functionalities, were considered helpful. One participant additionally raised the concern that automatically generated models may bypass the user's need to fully understand the underlying document, potentially introducing bias. Hence, the fine-grained traceability, as suggested by the participant, could help mitigate this problem. Another participant again pointed out the need for the AI-assisted document search capability as suggested.

Discussion

This chapter discusses the evaluation's findings and reflects on their implications for system design.

Usability and User Experience

Among these usability issues, some can be mitigated and improved through design, while others are more challenging to be improved for the system.

The first finding is that document readability is key to the tool's design, as it aims to enable users to start using it by reading. If a user doesn't have a good reading experience, turns to reading the document in their familiar way, and then finds the same text position within the tool, it will significantly affect the tool's designed usage scenario and reduce efficiency.

For the interaction design, some of the improvement ideas have been made up; e.g., to avoid the unnecessary review step, we can easily offer a project settings feature. And for the transparent AI input or system logic, it can be improved by showing instructions, e.g., using a switch button to give the user more flexible control over the AI input. In contrast, as this tool introduces a new way of process modelling involving multiple documents and multiple models, e.g., deselecting the deselect current model to start a new model step, is quite representative, without coming up with a good solution, therefore, it requires users to be familiar with the new interaction and workflow, which raises the challenge for the system.

User Expectations

During the evaluation, participants raised more advanced expectations, which are out of the current system's scope, but are worth further discussion.

Firstly, the expected fine-grained traceability between text and model represents an advanced requirement that goes beyond the capabilities of current LLM-based process modelling tools. However, it is a particularly important feature in the context of document-based process modelling, as the system is designed to facilitate a easier verification for the traceability of the AI-generated model. Therefore, exploring approaches to enable fine-grained traceability can be considered a valuable direction for future work.

Secondly, the expectation of integrated AI-powered document intelligence is another advanced requirement that has not yet been included in the current system. However, it represents a promising and valuable direction for future development, as such capabilities could further enhance the system's efficiency and perceived usefulness. In particular, enabling intelligent document understanding, contextual retrieval, and semantic linking between textual content and modelling elements could significantly streamline the modelling workflow and improve the overall user experience.

Limitations

Due to the small sample size (n=4), all of whom are modelling experts, the study did not include the perspective of domain experts. Besides, although the material provided is a real-world SOP document, it is a really well-written one with appropriate structure and not too long, 5 pages. So the document readability issue could affect the tool's usability more.

Besides, the evaluation didn't mention it as a limitation that the system is currently designed so that a model can be generated from a single document, without enabling cross-document modeling, which may require a more complex user interaction design.

Conclusion

This work is motivated by the inconvenience of state-of-the-art AI-assisted process modeling tools, which typically only provide a text input box and lack of support for document-based sources.

To resolve this issue, we designed, implemented, and evaluated an integrated web-based process modeling tool[4], especially for the document-based process modeling scenario, offering a seamless, efficient, and user-friendly workflow.

We first revisit the research questions:

RQ1 How do existing tools support AI-assisted process modelling with documents?

A thorough systematic review of existing tools and approaches for AI-assisted process modelling is conducted. The review shows that most of the existing tools still focus on model generation capabilities and quality. There is a trend to augment the input model with audio. However, the use case of creating process models from documents or a segment of documents remains missing.

RQ2: What functionalities should a system provide to support AI-assisted process modelling from documents, and how should these functionalities be designed and integrated in a seamless and user-friendly manner?

We derived minimal core requirements as follows: a unified workspace for documents and models; direct text selection from documents as LLM input; side-by-side visualization with explicit text–model traceability; and subprocess relationship management. Additional enhanced features include LLM-based model refinement, model versioning, document update support, bulk management, and interactive project graph visualization.

RQ3 How do users perceive the benefits and limitations of the system?

An evaluation with four process modeling experts indicates that the system is generally intuitive and usable, with particular value for creating and exploring multiple models from documents, supported by features such as text–model links and process–subprocess links. The evaluation implies two valuable future work directions: fine-grained text–model traceability and AI-powered document intelligence, both of which have particular potential to improve efficiency and user experience.

Overall, this work demonstrates the feasibility and practical value of supporting document-based process modeling through the developed prototype. It introduces a novel input modality that enables

[4] https://github.com/ga94hor/doc-based-process-modeler

direct text selection from uploaded documents as LLM input. Future work will focus on further improving the system's usability, extending support to additional document types, and conducting research in fine-grained traceability and the integration of AI-powered document intelligence.

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Appendix